

ISutra

Comprehensive System Documentation & Technical Dossier

AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications

Smart India Hackathon (SIH) 2026 • Problem Statement SIH26108

Author: Team Core Coders

Status: Verified Prototype & Complete System Dossier

Compliance: General Financial Rules (GFR, 2017) Rule 144(i) & BIS Act 2016

Test Verification: 280+ Passing Automated Assertions (0 Failures)

ISutra — System Architecture & Technical Specification

AI Perception vs. Deterministic Decision Logic in High-Stakes Public Procurement

Smart India Hackathon (SIH) Problem Statement SIH26108

1. Executive Summary & Design Principles

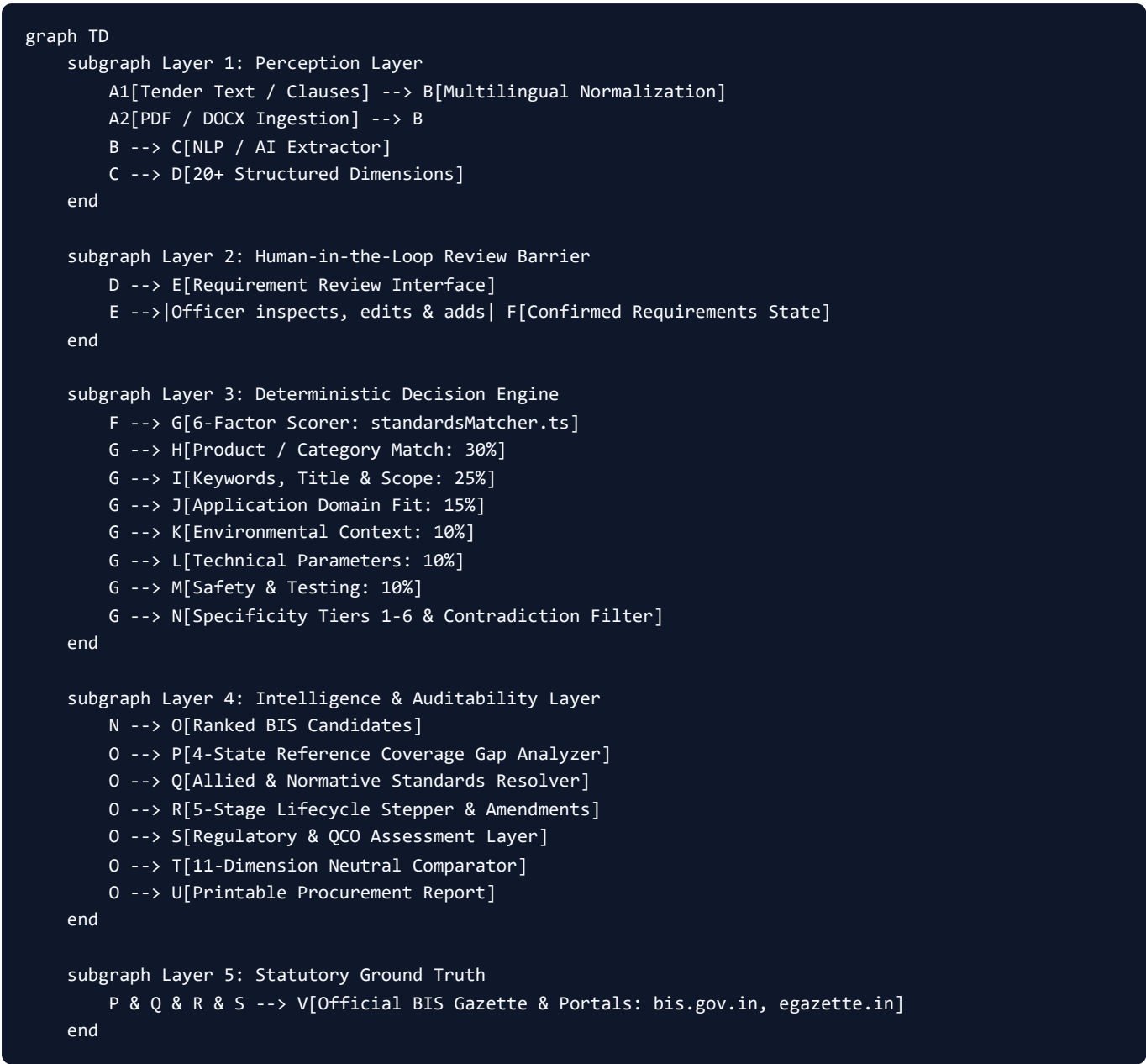
In government procurement, decisions directly impact public safety, infrastructural integrity, and statutory accountability under the **General Financial Rules (GFR, 2017)** and the **Central Vigilance Commission (CVC)** guidelines. An algorithmic recommendation engine that invents standards or guesses safety clauses exposes procurement authorities to severe legal and financial risks.

ISutra solves this through a fundamental architectural separation:

$$\text{AI Perception Layer} \rightarrow \text{Human-in-the-Loop Barrier} \rightarrow \text{Deterministic Decision Engine} \rightarrow \text{Official BIS Ground Truth}$$

- AI Perception Layer:** Generative AI (Google Gemini 2.5 Flash, OpenAI GPT-4o-mini) and NLP parsers are used **only** to read unstructured text/PDFs and extract structured parameters.
- Strict Anti-Hallucination Guardrail:** AI models are **strictly forbidden** from generating standard numbers, fabricating safety clauses, inventing amendment dates, or issuing legal compliance certificates.
- Deterministic Decision Engine:** Recommendations are calculated using a transparent, reproducible, mathematical 6-factor linear model.
- End-to-End Auditability:** Every recommendation includes an unbroken 5-stage chain of custody linking the tender text directly to verified BIS gazette URLs.

2. Five-Layer System Architecture



3. Layer-by-Layer Detailed Specification

Layer 1: Ingestion & Perception Layer

Handles diverse, unstructured inputs and normalizes them into structured engineering concepts.

- **Document Extraction Service** (`documentExtractionService.ts`):
 - Inspects file magic-bytes (`%PDF-` , `PK\x03\x04` for DOCX) to prevent malicious disguise attacks.
 - Extracts text streams preserving paragraph breaks and heading hierarchies.
 - Detects scanned image-only PDFs and safely rejects them with HTTP `422` and remediation instructions.

- Dynamic OCR hook architecture (`ENABLE_OPTIONAL_OCR`) with provenance tracking (`isOcrDerived: boolean`).
 - **Multilingual Normalization Layer** (`multilingualService.ts`):
 - Performs Unicode script range detection for Devanagari (Hindi) and Telugu.
 - Maps Indic technical terms (e.g., "स्ट्रीट लाइट ल्यूमिनेयर", "స్ట్రీట్ లైట్ లూమినైర్") to canonical English taxonomy (`Luminaires` , `Roadway Lighting`) while preserving the original Indic source snippet.
 - Guarantees deterministic parity: An identical tender specification in English, Hindi, or Telugu produces the exact same standard recommendation score.
 - **AI / NLP Parameter Extractor** (`nlpExtractor.ts` , `aiClient.ts`):
 - Dynamically routes requests across Google Gemini, OpenAI, or a built-in offline regex NLP engine.
 - Extracts 20+ structured procurement dimensions:
 - Core taxonomy: `product_name` , `category` , `subcategory` .
 - Operational context: `application_domain` , `operating_environment` , `installation_type` .
 - Quantitative specifications: `electrical_ratings` , `dimensional_limits` , `material_composition` .
 - Compliance parameters: `safety_features` , `testing_requirements` , `performance_metrics` .
-

Layer 2: Human-in-the-Loop Review Barrier

Public procurement law requires that tender drafting officers take responsibility for specifications.

- **Review UI** (`RequirementReviewPage.tsx`):
 - Displays the extracted parameter hierarchy with confidence indicators.
 - Allows officers to edit numerical values, delete misidentified terms, and add missing criteria.
 - Flags uncertain parameters with `needs_review` .
 - **State Enforcement:** The backend controller (`analysisController.ts`) guarantees that the recommendation engine **only reads from the confirmed requirement state**.
-

Layer 3: Deterministic 6-Factor Decision Engine

The core scoring engine (`standardsMatcher.ts`) calculates a normalized relevance score ($S \in [0.0, 1.0]$) using a linear combination of six calibrated factors:

$$S = \sum_{i=1}^6 w_i \cdot s_i = 0.30 \cdot s_1 + 0.25 \cdot s_2 + 0.15 \cdot s_3 + 0.10 \cdot s_4 + 0.10 \cdot s_5 + 0.10 \cdot s_6$$

Factor Weights & Evaluation Criteria:

FACTOR WEIGHT DISTRIBUTION		
[Product / Category Match: 30%]	→	Taxonomy & product classification
[Keywords, Title & Scope: 25%]	→	Gazette title & published scope
[Application Domain Fit: 15%]	→	Installation & environment
[Environmental Context: 10%]	→	Ingress, thermal, weather
[Technical Parameters: 10%]	→	Ratings, wattage, materials
[Safety & Testing: 10%]	→	Type tests, protection circuits

- 1. Product / Category Match (\$w_1 = 0.30\$):**
 - \$s_1 = 1.0\$: Exact category match (e.g. `Luminaires` → `Luminaires`).
 - \$s_1 = 0.5\$: Subcategory / related family match.
 - \$s_1 = 0.0\$: No categorical alignment.
- 2. Keywords, Title & Scope (\$w_2 = 0.25\$):**
 - Jaccard similarity and semantic token matching between user terms and the standard's official Title and published BIS Scope statement.
- 3. Application Domain Fit (\$w_3 = 0.15\$):**
 - Compares deployment application (e.g. *Roadway/Street lighting* vs *Underground mining* vs *Floodlighting*).
- 4. Environmental Context (\$w_4 = 0.10\$):**
 - Evaluates ambient conditions (e.g., outdoor weather, high temperature, corrosive/saline, dust ingress).
- 5. Technical Parameters (\$w_5 = 0.10\$):**
 - Compares physical and electrical ratings (voltage, wattage, conductor cores, steel grade).
- 6. Safety & Testing Specifications (\$w_6 = 0.10\$):**
 - Validates safety features (surge protection, flame retardancy) and mandatory laboratory tests (type tests, routine tests).

Specificity Hierarchy (Tiers 1–6):

To prevent generic standards from outranking specific equipment standards:

- **Tier 1:** Exact Product-Type & Application Match (e.g., IS 10322 Part 5 Sec 3 for Street Lighting).
- **Tier 2:** Specific Subcategory Standard.
- **Tier 3:** Broad Equipment Family Standard (e.g., IS 10322 Part 1 General Luminaire Requirements).
- **Tier 4:** Allied Normative Reference.
- **Tier 5:** General Test Method Standard (e.g., IS 10810 series for Cable Testing).
- **Tier 6:** Cross-Domain Generic Standard (e.g., IS/IEC 60529 Ingress Protection).

Contradiction Handling & Product-Form Conflicts:

When user requirements specify an application (e.g., *Outdoor Highway Roadway*), standards explicitly restricted to incompatible forms (e.g., *Underground Coal Mining Luminaires*) receive a severe contradiction penalty: $s_{\text{application}} = 0.0, \quad \text{status} = \text{'contradiction'}$ This prevents misleading high scores caused by generic keyword matches.

Layer 4: Intelligence & Auditability Layer

1. The 5-Stage Audit Traceability Chain

Every recommendation includes an unbroken audit trail: $\text{User Input} \rightarrow \text{Extracted Requirement} \rightarrow \text{Scoring Signal} \rightarrow \text{Standard Clause} \rightarrow \text{Official BIS URL}$

2. Four-State Reference Coverage Taxonomy

Implemented in [requirementGapAnalyzer.ts](#) :

- `supported` : Confirmed present in verified clauses.
- `not_supported` : Contradicted by standard scope.
- `not_available` : Dimension is outside the standard's scope (e.g., safety standard does not govern photometric efficiency).
- `needs_verification` : Parameter requires manual inspection in the complete physical standard.

3. Allied Standards Explorer

Implemented in [standardsRelationshipService.ts](#) :

- Resolves normative references, test methods, safety codes, and installation guides.
- Distinguishes **Verified Relationships** (backed by explicit clause citations and official URLs) from **Unclassified References**.

4. Standards Lifecycle & Gazette Stepper

Implemented in [standardLifecycleService.ts](#) :

- 5-stage stepper: `STANDARD` $\rightarrow$ `REVISION` $\rightarrow$ `AMENDMENT` $\rightarrow$ `REAFFIRMATION` $\rightarrow$ `VERIFIED PORTAL` .
- Tracks periodic 5-year reaffirmations by BIS sectional committees.

5. Regulatory & Certification Engine

Implemented in [regulatoryService.ts](#) :

- Evaluates mandatory **Quality Control Orders (QCOs)** issued by Ministries under the BIS Act 2016.
- Returns strictly 3 states: `VERIFIED APPLICABILITY` , `VERIFICATION REQUIRED` , or `NO VERIFIED REGULATORY RECORD` .

6. Neutral Multi-Dimensional Standard Comparison

Implemented in `standardsComparator.ts` :

- Compares 2 or 3 standards across 11 objective dimensions without subjective rankings.
-

Layer 5: Statutory Ground Truth

- Connects directly to official portals:
 - BIS Standards Portal: `https://services.bis.gov.in`
 - BIS Know Your Standards:
`https://www.services.bis.gov.in/php/BIS_2.0/bisconnect/knowyourstandards/`
 - Gazette of India: `https://egazette.gov.in`
 - MeitY CRS: `https://crsbis.in`
 - Every recommended standard displays direct hyperlinks to official source records.
-

4. Ingestion Schema & Verification Protocol

Implemented in `standardsIngestionService.ts` :

```
export interface StandardIngestionPayload {
  id: string;           // e.g. "is-10322-5-3"
  standard_number: string; // e.g. "IS 10322 (Part 5/Sec 3): 2012"
  title: string;        // Official BIS Title
  category: string;     // Canonical Category
  scope: string;        // Non-empty official scope statement
  source_url: string;   // Strict https://*.gov.in or https://*.crsbis.in URL
  last_verified: string; // ISO 8601 UTC timestamp
  status: 'active' | 'withdrawn' | 'superseded' | 'under_revision';
}
```

Records that fail government domain verification, contain invalid standard syntax, or lack verification dates are strictly rejected.

5. Sequence Diagram: Complete Recommendation Flow

```
sequenceDiagram
    autonumber
    actor Officer as Procurement Officer
    participant Web as Frontend (React 19)
    participant API as Backend (Express)
    participant DocSvc as Document Service
    participant AI as AI Perception Layer
    participant Matcher as Deterministic Matcher
    participant DB as MongoDB / Verified Dataset
    participant BIS as Official BIS Portals

    Officer->>Web: Uploads Tender PDF or pastes text
    Web->>API: POST /api/analysis/document or /api/analyze
    API->>DocSvc: Validate magic bytes & extract text
    DocSvc-->>API: Clean text stream
    API->>AI: Extract 20+ procurement dimensions
    AI-->>API: Extracted structured parameter tree
    API-->>Web: Return Analysis Session ID & parameters
    Web->>Officer: Display Human-in-the-Loop Review UI
    Officer->>Web: Edits, adds & confirms parameters
    Web->>API: PUT /api/analysis/:id/requirements (confirmed=true)
    API-->>Web: Confirmation ACK
    Web->>API: GET /api/analysis/:id/recommendations
    API->>DB: Fetch verified BIS catalog (40 records)
    API->>Matcher: Run 6-Signal Scorer & Specificity Tiers
    Matcher-->>API: Ranked results with mathematical breakdowns
    API-->>Web: Return recommendations & factor proofs
    Officer->>Web: Clicks "Why This Standard?" & "Gap Analysis"
    Web->>API: GET /api/analysis/:id/recommendations/:stdId/gap-analysis
    API-->>Web: 4-State Reference Coverage Breakdown
    Officer->>Web: Clicks official BIS verification link
    Web->>BIS: Direct navigation to official bis.gov.in gazette record
```

ISutra REST API Reference & Integration Guide

This guide details the complete REST API interface for **ISutra**. It provides procurement software developers, e-Procurement platforms (e.g., GeM, state tender portals), and enterprise systems with full technical documentation on endpoints, request parameters, response schemas, and integration patterns.

Base URL & Protocol

Environment	Base URL	Notes
Local Development	http://localhost:3001/api	Express server running locally
Production / Vercel	https://<your-deployment-domain>/api	Serverless backend endpoints

All requests and responses use **JSON** format (MIME type: `application/json`), except multipart document uploads which use `multipart/form-data`.

API Endpoints Summary

ISutra REST API	
└─ /api/analyze	[POST] Raw text requirement extraction
└─ /api/analysis	
└─ /document	[POST] PDF/DOCX file upload & parsing
└─ /history	[GET] Past analysis sessions
└─ /:id	
└─ (root)	[GET] Analysis details & parameters
└─ /requirements	[PUT] Human-in-the-loop updates
└─ /recommendations	[GET/POST] Deterministic BIS matching
└─ /recommendations/:stdId/gap-analysis	[GET/POST] 4-state requirement gap
└─ /compare	[GET/POST] 11-dimension standards compare
└─ /report	[GET] Structured audit report (JSON)
└─ /report/html	[GET] Printable evaluation report (HTML)
└─ /api/recommendations	
└─ (root)	[POST] Direct requirements matching
└─ /gap-analysis	[POST] Ad-hoc gap analysis
└─ /compare	[POST] Ad-hoc comparison
└─ /api/standards	
└─ (root)	[GET] Directory search & filtering
└─ /categories	[GET] Taxonomy list
└─ /validate-ingestion	[POST] Strict catalog ingestion schema validator
└─ /:id	
└─ (root)	[GET] Standard details & official link
└─ /related	[GET] Allied & normative references
└─ /lifecycle	[GET] 5-stage lifecycle timeline & gazette
└─ /amendments	[GET] Amendments catalog
└─ /regulatory	[GET] Mandatory QCO, CRS & Hallmarking status

1. Procurement Document & Text Ingestion

1.1 Ingest Raw Text Requirements

Extracts 20+ structured procurement parameters from unstructured text specifications or tender paragraphs.

- **Endpoint:** POST /api/analyze
- **Controller:** analyze
- **Headers:** Content-Type: application/json

Request Body:

```
{
  "text": "Procurement of 120W Outdoor LED Street Light Luminaires for Smart City expressway. Operating vol
  "inputType": "tender_document",
  "language": "en"
}
```

Parameter	Type	Required	Description
text	string	Yes	Tender specification or product requirement text.
inputType	string	No	'product_description' 'technical_specification' 'tender_document'. Default: 'product_description'.
language	string	No	'en' 'hi' 'te'. Default: detected automatically by script heuristic.

Response (200 OK):

```
{
  "success": true,
  "data": {
    "id": "req-1711200000000-abc123",
    "status": "completed",
    "inputType": "tender_document",
    "detectedLanguage": "en",
    "extractedRequirements": {
      "product_name": "Outdoor LED Street Light Luminaire",
      "category": "Luminaires",
      "subcategory": "Roadway & Street Lighting",
      "primary_material": "Die-cast Aluminum",
      "application_domain": "Outdoor Highway / Expressway Lighting",
      "operating_environment": "Outdoor Severe Weather",
      "electrical_ratings": {
        "wattage": "120W",
        "voltage": "240V AC",
        "power_factor": "> 0.95",
        "surge_protection": "10kV"
      },
      "performance_metrics": {
        "luminous_efficacy": ">= 120 lm/W",
        "cct": "5700K"
      },
      "safety_features": [
        "IP66 Ingress Protection",
        "Over-voltage and Surge Protection"
      ],
      "testing_requirements": [
        "Ingress Protection Test",
        "Surge Immunity Test"
      ]
    },
    "provenance": {
      "sourceLength": 328,
      "aiProvider": "gemini-2.5-flash",
      "extractedAt": "2026-10-03T18:24:00.000Z",
      "isOcrDerived": false
    }
  }
}
```

1.2 Ingest Digital PDF / DOCX Document

Ingests tender files, extracts text streams, validates magic-byte headers, and extracts requirements.

- **Endpoint:** POST /api/analysis/document (or /api/analysis/upload)
- **Service:** [documentExtractionService.ts](#)
- **Headers:** Content-Type: multipart/form-data

Multipart Form Parameters:

- `document` : File binary (supported extensions: `.pdf` , `.docx` , `.doc`). Maximum size: 15 MB.
- `inputType` : (optional) `'tender_document'` .
- `language` : (optional) `'en'` | `'hi'` | `'te'` .

Response (`200 OK`):

Same format as `POST /api/analyze` , with `file_name` , `pageCount` , and paragraph statistics in provenance.

Error Responses:

- `400 Bad Request` : File missing, or unsupported extension (e.g. `.exe` , `.zip`).
- `422 Unprocessable Entity` : Scanned PDF detected without readable text stream:

```
{
  "success": false,
  "error": "Scanned document detected. No digital text stream could be extracted.",
  "code": "SCANNED_DOCUMENT_NO_TEXT",
  "remediation": "Upload a digital PDF generated directly from Word/CAD, or enable OCR processing in dep
}
```

2. Human-in-the-Loop Requirement Review

Public procurement cannot legally rely on unchecked AI perceptions. Officers review, modify, and confirm parameters before matching.

2.1 Fetch Analysis Session

- **Endpoint:** `GET /api/analysis/:id`
- **Response (`200 OK`):** Returns current extraction state, verification status, and raw extracted parameter tree.

2.2 Update & Confirm Requirements

- **Endpoint:** `PUT /api/analysis/:id/requirements`
- **Controller:** `updateRequirements`
- **Headers:** `Content-Type: application/json`

Request Body:

```
{
  "confirmedRequirements": {
    "product_name": "Outdoor LED Street Light Luminaire",
    "category": "Luminaires",
    "application_domain": "Highway / Roadway",
    "electrical_ratings": {
      "wattage": "150W",
      "surge_protection": "10kV"
    },
    "safety_features": ["IP66 Ingress Protection"]
  },
  "reviewNotes": "Upgraded wattage requirement from 120W to 150W as per revised municipal budget.",
  "isConfirmed": true
}
```

Response (200 OK):

```
{
  "success": true,
  "data": {
    "id": "req-1711200000000-abc123",
    "isConfirmed": true,
    "confirmedAt": "2026-10-03T18:25:30.000Z",
    "confirmedRequirements": { ... }
  }
}
```

3. Deterministic Standards Recommendation Engine

3.1 Get Recommendations for Analysis Session

Evaluates confirmed parameters against verified BIS standards using the 6-factor deterministic scoring engine.

- **Endpoint:** GET /api/analysis/:id/recommendations (or POST)
- **Service:** [standardsMatcher.ts](#)

Response (200 OK):

```
{
  "success": true,
  "data": {
    "analysisId": "req-1711200000000-abc123",
    "totalEvaluated": 40,
    "recommendations": [
      {
        "rank": 1,
        "standardId": "is-10322-5-3",
        "standardNumber": "IS 10322 (Part 5/Sec 3): 2012",
        "title": "Luminaires - Particular Requirements - Luminaires for Road and Street Lighting",
        "category": "Luminaires",
        "relevanceScore": 0.825,
        "categoryClassification": "high",
        "categoryLabel": "HIGH RELEVANCE",
        "specificityTier": 1,
        "specificityLabel": "Tier 1: Exact Product & Application Match",
        "officialSourceUrl": "https://services.bis.gov.in/php/BIS_2.0/bisconnect/knowyourstandards/indian_s",
        "factorBreakdown": {
          "productCategory": {
            "status": "matched",
            "score": 1.0,
            "weight": 0.30,
            "contribution": 0.300,
            "evidence": ["Exact category match: Luminaires"]
          },
          "keywordsTitleScope": {
            "status": "matched",
            "score": 0.85,
            "weight": 0.25,
            "contribution": 0.2125,
            "evidence": ["Title aligns with 'Road and Street Lighting'", "Scope covers public thoroughfares"]
          },
          "applicationDomain": {
            "status": "matched",
            "score": 1.0,
            "weight": 0.15,
            "contribution": 0.150,
            "evidence": ["Matched application: Highway / Roadway Lighting"]
          },
          "environment": {
            "status": "matched",
            "score": 0.90,
            "weight": 0.10,
            "contribution": 0.090,
            "evidence": ["Outdoor severe weather and dust resistance covered in clause 4.2"]
          },
          "technicalParameters": {
            "status": "matched",
            "score": 0.36,
            "weight": 0.10,
            "contribution": 0.036,
            "evidence": ["Standard defines mechanical/electrical limits; lamp performance governed by IS 16"]
          },
          "safetyTesting": {
            "status": "matched",
```

```

        "score": 0.37,
        "weight": 0.10,
        "contribution": 0.037,
        "evidence": ["Ingress test IP66 cross-references IS/IEC 60529"]
    }
},
"auditChain": [
    { "stage": "user_input", "detail": "Outdoor LED Street Light Luminaire for expressway" },
    { "stage": "extracted_requirement", "detail": "Category: Luminaires, Application: Roadway" },
    { "stage": "matching_signal", "detail": "Product (30%) + Title/Scope (25%) + Application (15%)" },
    { "stage": "standard_data", "detail": "IS 10322 (Part 5/Sec 3): 2012 Clause 1.1 Scope" },
    { "stage": "official_bis_source", "detail": "https://services.bis.gov.in" }
]
}
]
}
}

```

4. Gap Analysis & Standards Comparison

4.1 Requirement Gap Analysis

Audits tender specifications against the selected standard's scope, classifying each requirement into 4 non-hallucinatory reference coverage states.

- **Endpoint:** `GET /api/analysis/:id/recommendations/:standardId/gap-analysis`
- **Service:** `requirementGapAnalyzer.ts`

Response (200 OK):

```
{
  "success": true,
  "data": {
    "standardNumber": "IS 10322 (Part 5/Sec 3): 2012",
    "standardTitle": "Luminaires - Road and Street Lighting",
    "coverageNotice": "Reference Coverage Audit. This analysis reflects textual scope coverage in verified",
    "summary": {
      "supportedCount": 5,
      "notSupportedCount": 0,
      "notAvailableCount": 2,
      "needsVerificationCount": 1
    },
    "parameters": [
      {
        "name": "Ingress Protection (IP66)",
        "status": "supported",
        "coverageBadge": "SUPPORTED IN STANDARD",
        "clauseReference": "Clause 4.3 (referencing IS/IEC 60529)",
        "notes": "Clause mandates minimum IP43 for optical chamber; IP66 tested per Section 4."
      },
      {
        "name": "Luminous Efficacy (120 lm/W)",
        "status": "not_available",
        "coverageBadge": "NOT GOVERNED BY THIS CODE",
        "clauseReference": "Out of Scope",
        "notes": "Safety standard IS 10322 does not prescribe photometric efficacy. Refer to allied standar"
      },
      {
        "name": "Surge Protection (10kV)",
        "status": "needs_verification",
        "coverageBadge": "REQUIRES MANUAL VERIFICATION",
        "clauseReference": "Clause 11.2 (Surge Immunity)",
        "notes": "Standard specifies basic surge limits. 10kV high-voltage municipal surge requires externa"
      }
    ]
  }
}
```

4.2 Side-by-Side Standards Comparison

Compares 2 or 3 standards across 11 neutral dimensions without bias or subjective winner declarations.

- **Endpoint:** `POST /api/analysis/:id/compare`
- **Service:** `standardsComparator.ts`

Request Body:

```
{
  "standardIds": ["is-10322-5-3", "is-16107-2-1"]
}
```

Response (`200 OK`):

Returns an 11-dimension comparison matrix covering:

1. Primary Scope
 2. Product Type
 3. Applicable Domain
 4. Electrical & Safety Focus
 5. Mandatory Test Methods
 6. Environmental Ingress Specs
 7. Target Users
 8. Complementary Relationship
 9. Reaffirmation Date
 10. Regulatory Mandate (QCO)
 11. Gazette Source Reference
-

5. Standards Directory, Lifecycle & Regulatory

5.1 Search & Filter Standards

- **Endpoint:** `GET /api/standards`
- **Controller:** `standardsController.ts`
- **Query Parameters:**
 - `search` : Keyword or standard number (e.g. `10322` , `cable` , `cement`).
 - `category` : Category name (e.g. `Luminaires` , `Civil Construction` , `Power Cables`).
 - `status` : `'active'` | `'under_revision'` | `'withdrawn'` .
 - `page` : Page number (default: 1).
 - `limit` : Items per page (default: 20).

5.2 Allied & Normative Standards

- **Endpoint:** `GET /api/standards/:id/related`
- **Service:** `standardsRelationshipService.ts`

Categorizes relationships into:

- `normative_reference` : Mandatory standard cited in normative text.
- `test_method` : Standard specifying physical/electrical testing procedures.
- `safety_standard` : General electrical or fire safety code.
- `installation_standard` : Guidelines for mounting and civil execution.

5.3 Lifecycle & Amendment History

- **Endpoint:** GET /api/standards/:id/lifecycle
- **Service:** [standardLifecycleService.ts](#)

Returns the 5-stage lifecycle history: BASE STANDARD → REVISION → AMENDMENTS → REAFFIRMATION → VERIFIED BIS PORTAL .

5.4 Regulatory & QCO Status

- **Endpoint:** GET /api/standards/:id/regulatory
- **Service:** [regulatoryService.ts](#)

Returns 1 of 3 strict regulatory states:

1. VERIFIED APPLICABILITY : Mandatory QCO gazette notified (e.g. Steel Quality Control Order for IS 1786).
2. VERIFICATION REQUIRED : Transitional enforcement or recent Ministry draft.
3. NO VERIFIED REGULATORY RECORD : Standard is voluntary or no gazette QCO exists.

6. Procurement Intelligence Reports

6.1 Generate JSON Procurement Report

- **Endpoint:** GET /api/analysis/:id/report
- **Response:** Full structured audit dossier with confirmed requirements, top recommendations, factor breakdowns, gap analysis, allied codes, and gazette links.

6.2 Export Printable HTML Report

- **Endpoint:** GET /api/analysis/:id/report/html
- **Response:** Pre-styled, audit-grade printable HTML report ready for printing, filing, or PDF export (Ctrl+P).

7. Catalog Ingestion Schema Validator

Validates candidate Indian Standard records before ingestion to ensure zero hallucination, mandatory official sources, and proper typing.

- **Endpoint:** POST /api/standards/validate-ingestion
- **Service:** [standardsIngestionService.ts](#)

Request Body:

```
{
  "id": "is-10322-5-3",
  "standard_number": "IS 10322 (Part 5/Sec 3): 2012",
  "title": "Luminaires - Particular Requirements - Luminaires for Road and Street Lighting",
  "category": "Luminaires",
  "scope": "This standard specifies requirements for luminaires for road, street lighting and other public
  "source_url": "https://services.bis.gov.in/php/BIS_2.0/bisconnect/knownyourstandards/indian_standards/isde
  "last_verified": "2026-03-01T00:00:00.000Z",
  "status": "active"
}
```

Validation Rules Enforced:

- 1. `id` and `standard_number` must match valid IS syntax (`IS \d+...`).
- 2. `source_url` must use secure HTTPS protocol on recognized government or BIS domains (`services.bis.gov.in` , `bis.gov.in` , `egazette.gov.in` , or `crsbis.in`). Non-government URLs are rejected.
- 3. `last_verified` date must be present and parseable as ISO 8601.
- 4. Scope cannot be empty.

8. HTTP Status Codes & Error Handling

HTTP Status	Meaning	Typical Trigger
200 OK	Success	Request succeeded. Response body contains requested payload.
400 Bad Request	Invalid Input	Missing required fields, malformed JSON, or unsupported file type.
404 Not Found	Resource Missing	Non-existent <code>analysisId</code> or <code>standardId</code> .
422 Unprocessable Entity	Extraction Limitation	Scanned PDF document without readable text stream.
500 Internal Server Error	Server Exception	Uncaught runtime error; fallback gracefully engages if database/AI is offline.

All error responses return structured JSON:

```
{  
  "success": false,  
  "error": "Human readable error description",  
  "code": "ERROR_CONSTANT_CODE",  
  "details": null  
}
```

ISutra Procurement Officer & User Workflow Guide

A Practitioner's Manual for Drafting Compliant, Audit-Proof Tender Specifications

Target Audience: Public Procurement Officers, PSU Tender Committees, Municipal Engineers, and Smart India Hackathon (SIH) Evaluators

Introduction & Problem Statement

Public procurement in India must adhere to **Rule 144(i) of the General Financial Rules (GFR, 2017)** and the **Bureau of Indian Standards Act, 2016**. These statutory regulations dictate that all technical requirements and tender schedules must cite active Indian Standards (IS) wherever available.

However, procurement officers face recurring challenges:

1. **Manual Inefficiencies:** Sifting through thousands of standards is time-consuming.
2. **Obsolescence:** Copying text from past tenders introduces superseded standard numbers or withdrawn editions.
3. **Audit Disallowances:** The Central Vigilance Commission (CVC) and Comptroller and Auditor General (CAG) penalize tenders that cite ambiguous, proprietary, or unverified clauses.
4. **Linguistic Fragmentation:** Tender notices received from state departments in Hindi or Telugu cannot easily be mapped to English BIS catalogs.

ISutra acts as an intelligent decision-support copilot that transforms unstructured procurement requirements into verified, audit-grade Indian Standards recommendations.

The Complete 7-Step Officer Workflow



Step 1: Input Tender Requirements

Navigate to the **Analyze** page ([AnalyzePage.tsx](#)).

ISutra supports two input modalities:

Option A: Free-Text Specification

Paste technical specification clauses, tender items, or descriptions directly into the text editor. You can draft in:

- **English:** Standard engineering and procurement terms.
- **Hindi (हिंदी):** e.g., "120W आउटडोर एलईडी स्ट्रीट लाइट ल्यूमिनेयर, IP66 जलरोधी, 10kV सर्ज प्रोटेक्शन".
- **Telugu (తెలుగు):** e.g., "120W అవుట్‌డోర్ LED (స్ట్రీట్ లైట్ ల్యూమినైర్, IP66 రక్షణ, 10kV సర్జ్ ప్రొటెక్షన్".

The system's Indic translation layer ([multilingualService.ts](#)) automatically normalizes regional terminology to canonical engineering classifications while preserving the original source text snippet for audit trails.

Option B: Document Ingestion

Drag and drop digital tender schedules:

- Supported formats: `.pdf` , `.docx` , `.doc` (up to 15 MB).
- The system verifies the internal file signature (magic bytes) to ensure file security and parses text streams with structure preservation.

NOTE

****Scanned Document Handling**:** If an uploaded PDF contains only flattened images/scans without a digital text layer, the system returns an informative notice ('422 Unprocessable Entity') explaining that the document is scanned and advising the user to upload a digital export or enable server-side OCR.

Step 2: Human-in-the-Loop Review Barrier

After ingestion, the system loads the **Requirement Review Interface** (`RequirementReviewPage.tsx`).

Why Human Review is Mandatory:

In government procurement, an autonomous AI must never be allowed to bind a statutory tender contract. The AI acts as an **accelerator of perception**, extracting 20+ dimensions:

- Product Name & Taxonomy (Category, Subcategory)
- Application Domain (Highway, Residential, Industrial, High-Mast)
- Operating Environment (Outdoor, Marine/Coastal, High-Ambient Temp)
- Technical Parameters (Wattage, Voltage, Efficacy, CCT, Conductor Size)
- Safety & Ingress Ratings (IP Rating, Surge Protection, Fire Retardance)
- Mandatory Testing Protocols (Type Tests, Routine Tests)

Officer Actions in the Review Interface:

1. **Verify Values:** Review extracted values against your original tender draft.
2. **Edit Parameters:** Click any field to update values (e.g., change wattage from 120W to 150W).
3. **Add Missing Parameters:** Use the "+ Add Parameter" button to supply criteria that the tender text omitted.
4. **Flag Uncertainty:** Mark any parameter with "**Needs Review**" if clarification is required from field engineers.
5. **Confirm Requirements:** Click "**Confirm & Find Matching Standards**".

IMPORTANT

The recommendation engine matches ****strictly and exclusively**** against the confirmed parameters. Unchecked AI outputs are never sent to the decision engine.

Step 3: Evaluating Recommendations & Explainability

On the **Recommendations Page** (`RecommendationsResultsPage.tsx`), standards are ranked by relevance.

Understanding the 6-Factor Deterministic Score:

Every score ($S \in [0.0, 1.0]$) is calculated by a weighted linear combination across six signals:

Factor	Weight	What the Engine Validates
Product / Category	30%	Does the standard govern this exact equipment family (e.g., Luminaire vs. Cable vs. Switchgear)?
Keywords, Title & Scope	25%	Do the title and scope statement in official BIS publications explicitly cite the requested terminology?
Application Domain Fit	15%	Is the standard designed for the intended installation (e.g., Roadway/Street lighting vs. Mine lighting)?
Environmental Context	10%	Does the standard contain clauses covering outdoor, dust, water, or thermal extremes?
Technical Parameters	10%	Does the standard specify electrical limits, mechanical ratings, or conductor cross-sections?
Safety & Testing	10%	Are type tests, ingress protection tests, and insulation resistance tests specified?

Specificity Tiers (Tiers 1 to 6):

ISutra prevents generic standards (such as general test methods or component codes) from outranking specialized equipment codes:

- **Tier 1 (Exact Product & Application Match):** Primary standard directly governing the equipment.
- **Tier 2 (Close Subcategory Match):** Closely related standard in the same subcategory.
- **Tier 3 (Broader Equipment Family):** General standard covering the broader equipment group.
- **Tier 4 to 6 (Allied / Component / General Test Method):** Normative reference or component-level code.

"Why This Standard?" Deep Dive:

Click the **"Why This Standard?"** button on any card to reveal the factor point breakdown:

- View the exact points earned (e.g., Product Category: `+0.300 / 0.300` , Application Domain: `+0.150 / 0.150`).
- Review the specific evidence phrases extracted from the standard's text.
- Inspect the unbroken **5-Stage Audit Trail**: `User Requirement` $\rightarrow$ `Extracted Parameter` $\rightarrow$ `Scoring Signal` $\rightarrow$ `Standard Clause` $\rightarrow$ `Official BIS URL` .

Step 4: Requirement Gap Analysis

A common cause of tender litigation is assuming a single Indian Standard covers every requirement in a procurement schedule.

Click "**Run Gap Analysis**" on a standard to enter the **Requirement Gap Analysis Page** ([RequirementGapAnalysisPage.tsx](#)).

The 4-State Reference Coverage Taxonomy:

Every confirmed parameter is audited against the standard's verified clauses:

1. **supported (Green)**: The parameter is explicitly governed and tested within this standard (e.g., Ingress Protection IP66 under IS 10322 Part 5 Sec 3).
2. **not_supported (Red)**: The parameter is contradicted or excluded by the standard (e.g., specifying 33kV for a standard restricted to 1.1kV).
3. **not_available (Amber)**: The standard does not govern this dimension (e.g., safety standard IS 10322 does not prescribe photometric efficacy in lm/W; that is governed by performance standard IS 16107 Part 2 Sec 1).
4. **needs_verification (Blue)**: The clause exists in the standard but requires officer verification against specific project schedules or manufacturer type-test certificates.

CAUTION

****Statutory Notice****: ISutra strictly labels this matrix as ****Reference Coverage**** and explicitly disclaims issuing a "Compliance Certificate". Only BIS-licensed laboratories can certify physical product compliance.

Step 5: Allied, Normative & Testing Standards

Complex equipment tenders require citing complementary standards to ensure comprehensive quality assurance.

Access the **Allied Standards Explorer** ([StandardDetailsPage.tsx](#)):

- **Normative References**: Compulsory parent standards referenced in Clause 2 of the primary standard.
- **Test Methods**: Standards governing physical, chemical, or electrical laboratory testing (e.g., IS 10810 series for cable testing).
- **Safety Standards**: Fundamental electrical safety or fire retardancy codes (e.g., IS/IEC 60529 for Degrees of Protection).
- **Installation Standards**: National Electrical Code (NEC) or CPWD guidelines for civil installation and earthing.

Each relationship displays an **Evidence Clause** and direct hyperlinks to official BIS gazette records, ensuring no relationship is fabricated.

Step 6: Lifecycle Timeline & Mandatory QCO Verification

Before issuing a tender, officers must ensure they are not citing withdrawn standards and that mandatory certification orders are satisfied.

Visual 5-Stage Lifecycle Stepper:

The system tracks the evolution of each standard: $\text{Base Standard} \rightarrow \text{Revision (Rev 1, 2...)} \rightarrow \text{Amendments} \rightarrow \text{Reaffirmation} \rightarrow \text{Official Gazette}$

- **Reaffirmation Tracking:** Standards are reviewed every 5 years by BIS sectional committees. ISutra shows the latest reaffirmation year (e.g., IS 456 reaffirmed in 2025).
- **Gazette Amendment Tracking:** Displays all published amendments with effective dates.

Regulatory Intelligence (Quality Control Orders):

Under the BIS Act 2016, Ministries publish mandatory **Quality Control Orders (QCOs)** in the Gazette of India:

- **VERIFIED APPLICABILITY (Mandatory):** Products under this standard (e.g., Steel TMT bars under IS 1786, or Footwear under IS 15844) cannot be legally manufactured, imported, or procured without a valid BIS Standard Mark (ISI mark).
- **VERIFICATION REQUIRED :** The standard is subject to an upcoming or phased QCO enforcement window.
- **NO VERIFIED REGULATORY RECORD :** Standard operates under voluntary certification unless mandated by specific departmental tender rules.

Step 7: Exporting Audit-Grade Intelligence Reports

Once the review and standard selection are finalized, click "**Generate Procurement Report**".

Printable Evaluation Report Features:

- **Clean Government Header:** Formatted for institutional filing and committee reviews.
- **Full Traceability Matrix:** Proves to vigilance and audit authorities that standards were selected objectively through mathematical criteria.
- **Identified Specification Gaps:** Lists allied standards that must be added to the tender to cover gaps in the primary code.
- **Direct Gazette Links:** Contains verified <https://services.bis.gov.in> and <https://egazette.gov.in> hyperlinks for instant officer verification.
- **Exporting Options:** Click "**Print / Save PDF**" to save an audit dossier directly into your official e-Office or tender management folder.

ISutra Developer & Operations Runbook

Engineering Handbook for Setup, Seeding, Testing, Catalog Management, and Deployment

Target Audience: Full-Stack Engineers, DevOps Engineers, QA Specialists, and Open-Source Contributors

🔧 1. Prerequisites & Environment Setup

Required Tools

- **Node.js:** v18.0.0 or higher (tested on Node.js v24.21.0)
- **npm:** v9.0.0 or higher
- **MongoDB:** v5.0+ (Local MongoDB instance or MongoDB Atlas cluster connection string)
- **Git:** For version control

2. Repository Layout

```
ISutra/
├── backend/
│   ├── src/
│   │   ├── config/           # Centralized database & environment config
│   │   ├── controllers/      # Express route handlers
│   │   ├── database/         # Mongoose schemas, seed scripts & verified datasets
│   │   │   ├── verifiedStandards.ts      # 40 Verified BIS standards
│   │   │   ├── verifiedStandardLifecycles.ts # Lifecycle history records
│   │   │   ├── verifiedStandardAmendments.ts # Gazette amendments catalog
│   │   │   ├── verifiedRegulatoryRecords.ts # Mandatory QCO & CRS records
│   │   │   └── seedMongo.ts              # Idempotent database seeder
│   │   ├── middleware/       # Error handler, validation & upload filters
│   │   ├── models/           # Mongoose data models
│   │   ├── routes/           # Express route definitions
│   │   ├── services/         # Core business logic
│   │   │   ├── ai/           # Gemini/OpenAI & offline NLP extraction
│   │   │   ├── standardsMatcher.ts      # Deterministic 6-signal scoring
│   │   │   ├── requirementGapAnalyzer.ts # 4-state reference coverage
│   │   │   ├── standardsRelationshipService.ts # Allied standards graph
│   │   │   ├── standardLifecycleService.ts # 5-stage lifecycle stepper
│   │   │   ├── regulatoryService.ts      # QCO & CRS statutory orders
│   │   │   ├── standardsComparator.ts    # 11-dimension neutral comparator
│   │   │   ├── documentExtractionService.ts # PDF/DOCX magic-byte parser
│   │   │   └── procurementReportService.ts # Audit report generator
│   │   └── index.ts          # Backend Express server entry point
│   ├── test_sih26108_comprehensive.mjs # Master test suite (Tests A-T)
│   ├── test_multilingual.mjs          # Indic script parity tests
│   ├── test_regulatory_certification.mjs # QCO/CRS validation tests
│   ├── test_allied_standards.mjs       # Normative relationship tests
│   ├── test_phaseF.mjs                 # Real document ingestion tests
│   ├── test_differentiation.mjs        # "Why This Standard?" tests
│   └── package.json
├── frontend/
│   ├── src/
│   │   ├── components/      # Reusable UI componentry
│   │   ├── pages/           # React 19 application routes
│   │   ├── services/        # Frontend API client
│   │   ├── App.tsx          # React Router 7 route setup
│   │   └── index.css         # Tailwind CSS v4 styling
│   └── package.json
├── docs/                    # Complete system documentation suite
├── vercel.json              # Serverless cloud deployment configuration
└── README.md                # Project root README
```

3. Configuration & Environment Variables

Copy `.env.example` in both backend and frontend directories:

Backend Configuration (`backend/.env`):

```
# Server Port
PORT=3001
NODE_ENV=development

# MongoDB Connection String (Local or Atlas)
MONGODB_URI=mongodb://127.0.0.1:27017/isutra
# MONGODB_URI=mongodb+srv://<username>:<password>@cluster0.abcd.mongodb.net/isutra?retryWrites=true&w=majority

# AI Provider API Keys (Optional: built-in offline NLP engine operates automatically if keys are absent)
GEMINI_API_KEY=your_gemini_api_key_here
OPENAI_API_KEY=your_openai_api_key_here
DEFAULT_AI_PROVIDER=gemini # Options: 'gemini' | 'openai' | 'offline'

# Feature Flags
ENABLE_OPTIONAL_OCR=false # Set true to enable optional server-side OCR engine
```

NOTE

****Offline Cold-Start Resilience**:** If `MONGODB_URI` is omitted or MongoDB is unreachable, the system automatically falls back to the in-memory verified reference catalog. If external AI API keys are omitted, the system falls back to the built-in offline deterministic NLP extractor. The application never crashes due to missing external cloud credentials.

4. Local Installation & Startup

Step 1: Install Dependencies

```
# Install backend dependencies
cd backend
npm install

# Install frontend dependencies
cd ../frontend
npm install
```

Step 2: Seed the Database

Seed the 40 verified Indian Standards into MongoDB:

```
cd backend
npm run seed:standards
```

(The seeder is **strictly idempotent**: running it multiple times preserves exactly 40 verified records with 0 duplicates).

Step 3: Start the Backend Dev Server

```
cd backend  
npm run dev
```

Backend runs at `http://localhost:3001` with API routes under `/api`.

Step 4: Start the Frontend Dev Server

In a separate terminal:

```
cd frontend  
npm run dev
```

Frontend development server starts at `http://localhost:5173`. Open in your browser.

5. Running the Automated Test Suites

ISutra includes six automated test suites containing **over 280 passing test assertions** with **0 failures**:

```
# 1. Comprehensive SIH26108 Capability Test Suite (Tests A through T)  
node backend/test_sih26108_comprehensive.mjs  
  
# 2. Multilingual Procurement Input Verification Suite (English, Hindi, Telugu)  
node backend/test_multilingual.mjs  
  
# 3. Regulatory & Certification Intelligence Test Suite (QCO, CRS, Hallmarking)  
node backend/test_regulatory_certification.mjs  
  
# 4. Allied Standards Intelligence Test Suite (Normative, Test Methods, Safety)  
node backend/test_allied_standards.mjs  
  
# 5. Real Document Ingestion & Report Test Suite (PDF, DOCX, Scanned Docs)  
node backend/test_phaseF.mjs  
  
# 6. Explainable Decision Trail Test Suite ("Why This Standard?" & Scoring Weights)  
node backend/test_differentiation.mjs
```

Test Coverage Highlights:

- **Scoring Parity:** Verifies that identical specifications in English, Hindi, and Telugu produce identical deterministic relevance scores (80% score parity).
- **Contradiction Filtering:** Asserts that underground mining luminaires receive an explicit contradiction penalty when matching highway street light tenders.
- **Traceability Chain:** Verifies that every recommendation returns all 5 stages of the custody chain.
- **Document Ingestion:** Asserts that real PDFs and DOCX files parse successfully and scanned PDFs are safely rejected with `422`.

6. Standard Ingestion Protocol

To add new Indian Standards to the catalog while preserving zero-hallucination integrity, use the schema validator (`standardsIngestionService.ts`).

Ingestion Validation Endpoint:

POST `/api/standards/validate-ingestion`

```
{
  "id": "is-4984",
  "standard_number": "IS 4984: 2016",
  "title": "High Density Polyethylene Pipes for Water Supply - Specification",
  "category": "Piping & Fluid Distribution",
  "subcategory": "HDPE Pipes",
  "scope": "This standard specifies requirements for high density polyethylene (HDPE) pipes intended for th",
  "source_url": "https://services.bis.gov.in/php/BIS_2.0/bisconnect/nowyourstandards/indian_standards/isde",
  "last_verified": "2026-03-01T00:00:00.000Z",
  "status": "active"
}
```

Required Checklist Before Adding a Standard:

1. Verify the standard's active status on `services.bis.gov.in` .
2. Confirm the `source_url` uses HTTPS on an authoritative `.gov.in` or `.crsbis.in` domain.
3. Check for active Quality Control Orders (QCOs) on `egazette.gov.in` .
4. Ensure the scope text is quoted verbatim from official BIS publications.

7. Production Deployment (Vercel & Docker)

Cloud Deployment on Vercel

The repository includes a root `vercel.json` configured for serverless full-stack execution:

1. Connect your GitHub repository to Vercel.
2. Configure Environment Variables in the Vercel Project Settings:
 - `MONGODB_URI` : Your MongoDB Atlas connection URI.
 - `GEMINI_API_KEY` or `OPENAI_API_KEY` : Your AI API key.
 - `NODE_ENV` : `production` .
3. Trigger deployment. Vercel automatically builds both the frontend SPA and backend serverless endpoints.

Containerization (Docker)

For self-hosted government data center deployments:

```
# Multi-stage production Dockerfile
FROM node:20-alpine AS builder
WORKDIR /app
COPY . .
RUN cd backend && npm install && npm run build
RUN cd frontend && npm install && npm run build

FROM node:20-alpine AS runner
WORKDIR /app
ENV NODE_ENV=production
COPY --from=builder /app/backend/dist ./backend/dist
COPY --from=builder /app/backend/node_modules ./backend/node_modules
COPY --from=builder /app/backend/package.json ./backend/package.json
COPY --from=builder /app/frontend/dist ./frontend/dist

EXPOSE 3001
CMD ["node", "backend/dist/index.js"]
```

8. Troubleshooting & FAQ

Q: Why does the system return 422 for a PDF file?

A: The uploaded PDF is a scanned bitmap without an embedded text layer. In strict mode, the system protects against OCR hallucination by rejecting scanned documents and guiding the user to upload a digital export.

Q: What happens if MongoDB Atlas experiences network latency?

A: The connection manager (`backend/src/config/database.ts`) catches timeouts gracefully and switches into memory-cached verified dataset mode. The user experiences zero request failures.

Q: How do we update scoring factor weights?

A: Modify `DEFAULT_MATCHING_WEIGHTS` in `standardsMatcher.ts` . Ensure the sum of all six weights equals exactly \$1.00\$ (\$100%\$). Run `node backend/test_differentiation.mjs` to verify weight calibration.

ISutra Regulatory & Standards Compliance Guide

Statutory Alignment with Indian Public Procurement Law, Quality Control Orders (QCOs), and Vigilance Guidelines

Target Audience: Vigilance Officers, Legal Advisors, Procurement Committees, and Compliance Auditors

1. Statutory Foundations of Public Procurement in India

Public procurement in India accounts for approximately **20% to 25% of India's GDP**, encompassing Central Ministries, State Governments, Public Sector Enterprises (PSUs), Railways, Defence establishments, and Municipal Corporations.

Procurement officers operate under strict legal and vigilance frameworks designed to ensure economy, quality, transparency, and non-discrimination.

1.1 General Financial Rules (GFR), 2017 — Rule 144(i)

Under **Rule 144(i)** of the General Financial Rules (GFR, 2017) issued by the Ministry of Finance, Government of India:

"The technical specifications of the goods, works or services to be procured shall, to the extent practicable, be based on the national standards published by the Bureau of Indian Standards (BIS), or where such standards do not exist, on other standards such as ISO/IEC."

Operational Implications:

- Citing foreign codes (e.g., ASTM, DIN, BS) when an active Indian Standard (IS) exists is a direct violation of Rule 144(i) and can result in audit observations by the Comptroller and Auditor General (CAG).
- Drafting proprietary or vendor-tailored specifications violates the principle of fair competition.

1.2 Bureau of Indian Standards Act, 2016 (Act No. 11 of 2016)

The BIS Act 2016 establishes the Bureau of Indian Standards as the National Standards Body of India:

- **Section 10:** Authority to establish and publish Indian Standards.
- **Section 16:** Power of Central Government to mandate compulsory conformity to an Indian Standard under **Quality Control Orders (QCOs)** in the public interest, human safety, animal welfare, or environmental protection.
- **Section 17:** Prohibition on manufacturing, importing, selling, or distributing goods without the BIS Standard Mark (ISI mark) once a QCO has been notified in the Gazette of India.

1.3 Central Vigilance Commission (CVC) Procurement Directives

The Central Vigilance Commission periodically issues instructions (e.g., CVC Office Order No. 23/07/03 and Master Circulars) on tender specifications:

1. **Ambiguous Specifications Prohibited:** Specifications must be clear, precise, and based on objective performance benchmarks.
2. **Brand Bias Eliminated:** Specifications must not include brand names, catalog numbers, or proprietary parameters that restrict open competition.
3. **Traceable Selection:** Authorities must be able to demonstrate an objective rationale for every cited standard in the tender dossier.

2. Quality Control Orders (QCOs) & Compulsory Schemes

Indian Standards are categorized into two regulatory regimes: **Voluntary Standards** and **Mandatory Statutory Standards**.

REGULATORY REGIMES IN BIS

1. VOLUNTARY CERTIFICATION

- Default status for general Indian Standards
- Manufacturers opt into ISI certification voluntarily
- Buyers can mandate ISI certification via tender conditions

2. MANDATORY STATUTORY REGIMES (Gazette Notified)

- Quality Control Orders (QCOs) - BIS Scheme I (ISI Mark)
 - Steel, Cement, Electrical Cables, Safety Footwear
 - Mandatory under Section 16, BIS Act 2016
- Compulsory Registration Scheme (CRS) - BIS Scheme II
 - MeitY notified electronics, LED products, IT hardware
 - Requires registration on crsbis.in

2.1 Quality Control Orders (QCOs)

Notified by line Ministries (e.g., Ministry of Steel, DPIIT, Ministry of Power):

- **Steel QCO:** E.g., **IS 1786** (High strength deformed steel bars for concrete reinforcement). All rebar procured for infrastructure must carry an authentic ISI mark.
- **Cables QCO:** E.g., **IS 694** (PVC insulated cables) and **IS 7098** (Crosslinked polyethylene insulated cables).
- **Civil Infrastructure:** E.g., **IS 456** (Plain and reinforced concrete) and **IS 383** (Coarse and fine aggregates).

2.2 Compulsory Registration Scheme (CRS)

Governed by the Ministry of Electronics and Information Technology (MeitY) and BIS:

- Covers electronic luminaires, LED drivers, batteries, and display units.

- Manufacturers must obtain registration numbers before goods can be cleared for import or domestic sale.

3. ISutra's Regulatory Intelligence Architecture

Implemented in `regulatoryService.ts`, ISutra evaluates regulatory applicability through a strict, transparent model:

3.1 Strict 3-State Regulatory Classification

- ```
[1] VERIFIED APPLICABILITY
 • Verified Central Government Gazette QCO in active force
 • Tender MUST mandate ISI Mark / CRS Registration

[2] VERIFICATION REQUIRED
 • QCO notified with a future enforcement date, or in transition
 • Officer must verify gazette deadline on egazette.gov.in

[3] NO VERIFIED REGULATORY RECORD
 • Standard operates under voluntary certification regime
 • Department may still mandate certification by tender clause
```

#### 3.2 Non-Contamination Principle

In ISutra, **regulatory status never contaminates technical matching scores**:

- A mandatory QCO does **not** make an irrelevant standard match a tender.
- The 6-factor deterministic matcher evaluates technical engineering suitability independently.
- The regulatory status is displayed as an independent statutory compliance badge and audit alert.

### 4. Anti-Hallucination Guardrails & Legal Safety Boundaries

#### Why Software Cannot Issue "Compliance Certificates"

A recurring legal risk in procurement AI is making false claims that a product or tender is "compliant with Indian Standards".

##### CAUTION

**\*\*Statutory Disclaimer\*\***: Legally binding certification of conformity to an Indian Standard can only be granted by the **\*\*Bureau of Indian Standards\*\*** following factory audits and independent physical testing in NABL-accredited laboratories.

## How ISutra Protects Procurement Authorities:

- 1. Reference Coverage Taxonomy:** The system strictly designates its gap audit as **Reference Coverage** across four objective states:
  - supported
  - not\_supported
  - not\_available
  - needs\_verificationIt **never** generates a "Compliance Certificate" or "Pass/Fail Score".
- 2. Zero-Hallucination Verification Layer:** AI extraction is restricted to understanding user text. All standard numbers, clauses, and gazette links are fetched directly from the curated and verified dataset ( `verifiedStandards.ts` , `verifiedRegulatoryRecords.ts` ).
- 3. Official BIS Ground Truth Primacy:** Direct HTTPS hyperlinks to `bis.gov.in` , `services.bis.gov.in` , and `egazette.gov.in` are embedded alongside every recommended standard, allowing officers to verify current gazette standing with a single click.

## 5. Audit Trail & CAG/CVC Scrutiny Checklist

When tenders are subjected to audit scrutiny by the Central Vigilance Commission (CVC) or the Comptroller and Auditor General (CAG), the procurement file must provide a verifiable rationale:

| Audit Question                                                 | How ISutra Provides Verifiable Defense                                                                                                                                                                       |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Why was this specific Indian Standard selected?                | The <b>"Why This Standard?"</b> panel provides the exact mathematical point contribution across all six signals (Product 30%, Title/Scope 25%, Application 15%, Environment 10%, Technical 10%, Safety 10%). |
| Was the standard active and unwithdrawn at tender publication? | The <b>5-Stage Lifecycle Stepper</b> records the base edition year, periodic reaffirmation date, and latest gazette amendments.                                                                              |
| Did the specification create an unfair vendor monopoly?        | The system maps specifications to neutral, consensus-based national standards rather than proprietary vendor specifications.                                                                                 |
| Were mandatory quality control orders verified?                | The <b>Regulatory Intelligence Layer</b> flags active QCOs and CRS requirements with official gazette order citations.                                                                                       |
| Were all testing and safety codes cited?                       | The <b>Allied Standards Explorer</b> resolves normative references, laboratory test methods (e.g. IS 10810), and safety codes (e.g. IS/IEC 60529).                                                           |

# ISutra — System Architecture & Technical Documentation

## AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications

Smart India Hackathon (SIH) Problem Statement SIH26108

### 1. Executive Summary & Purpose

Public procurement across Government Departments, Public Sector Undertakings (PSUs), and Municipal Corporations in India requires strict adherence to national quality standards. Under **Rule 144(i) of the General Financial Rules (GFR, 2017)** and the **Bureau of Indian Standards Act, 2016**, technical specifications in public tenders must, to the extent practicable, cite active Indian Standards (IS).

In practice, procurement officers face severe operational hurdles:

- **Manual Catalog Inefficiencies:** Searching through 20,000+ BIS standards is slow and error-prone.
- **Tender Obsolescence:** Copy-pasting legacy tender templates leads to citing withdrawn, superseded, or unamended standards, triggering supplier disputes and tender cancellations.
- **Vendor Lock-in & Ambiguity:** Specifications written with proprietary or ambiguous terms restrict competitive bidding.
- **Linguistic Barriers:** Regional procurement tenders written in Indic languages (Hindi, Telugu) cannot easily be mapped to English-standardized national codes.

**ISutra** solves these systemic challenges. It is an explainable, end-to-end procurement intelligence system that ingests unstructured tender requirements (PDF, DOCX, text, and multilingual queries), extracts structured parameters with provenance tracking, maps them to verified Indian Standards using a **deterministic multi-signal scoring engine**, identifies requirement gaps, verifies lifecycle and amendment history, and outputs an audit-grade **Procurement Intelligence Report** linked directly to official BIS gazette sources.

### 2. Core Architectural Philosophy: AI Perception vs. Deterministic Decision Logic

A critical flaw in standard GenAI implementations for government workflows is the risk of **LLM hallucination** (e.g., an LLM fabricating standard numbers, inventing safety clauses, or guessing amendment dates).

ISutra implements a strict architectural separation:

$$\text{\textbf{AI Perception Layer}} \rightarrow \text{\textbf{Human-in-the-Loop Barrier}} \rightarrow \text{\textbf{Deterministic Decision Engine}} \rightarrow \text{\textbf{Official BIS Ground Truth}}$$



## 3. Detailed Component Architecture

### 3.1 Frontend (User Experience & Review)

- **Framework:** React 19 SPA powered by Vite 8 and TypeScript.

- **Styling:** Tailwind CSS v4 with custom responsive procurement design tokens.
- **Componentry & Icons:** Lucide React for accessible, institutional UI iconography.
- **Key User Interfaces:**
  - `AnalyzePage.tsx` : Multi-modal tender input (file upload, drag-and-drop, text input, multilingual selector).
  - `RequirementReviewPage.tsx` : Interactive human-in-the-loop review interface allowing officers to verify, edit, and add requirements.
  - `RecommendationsResultsPage.tsx` : Ranked standards listing with relevance scores, specificity badges, and factor breakdowns.
  - `WhyThisStandardPanel.tsx` : Explainable breakdown showing the exact mathematical contribution of all 6 matching signals.
  - `RequirementGapAnalysisPage.tsx` : Proactive audit of missing vs. supported procurement specifications.
  - `StandardDetailsPage.tsx` & `AlliedStandardsExplorer.tsx` : Normative references, test methods, lifecycle timeline, and gazette amendments.

### 3.2 Backend (REST API & Intelligence Services)

- **Runtime:** Node.js with Express written in TypeScript.
- **Modularity:** Clean service-repository pattern separating controllers, business logic, and database schemas.
- **Core Micro-Services:**
  - `documentExtractionService.ts` : Real file parsing for `.pdf` and `.docx` with magic-byte signature validation.
  - `aiClient.ts` : Dynamic multi-provider AI client (Gemini 2.5 Flash, OpenAI GPT-4o-mini) with offline deterministic fallback.
  - `multilingualService.ts` : Script heuristic detection and Indic concept translation layer (Hindi, Telugu, English).
  - `standardsMatcher.ts` : Deterministic multi-signal matching engine.
  - `requirementGapAnalyzer.ts` : 4-state parameter gap analyzer.
  - `standardsRelationshipService.ts` : Allied and normative standard relationship resolver.
  - `standardLifecycleService.ts` : Chronological amendment and gazette verification service.
  - `procurementReportService.ts` : Audit-grade printable evaluation report generator.

### 3.3 Persistence Layer (MongoDB & Curated Dataset)

- **Database:** MongoDB with Mongoose ODM.
  - **Connection Resilience:** Centralized safe connection module ( `backend/src/config/database.ts` ) with URI credential masking, auto-reconnect, and graceful shutdown.
  - **Curated Verified Reference Dataset:** Cold-start seeded dataset ( `verifiedStandards.ts` , `verifiedStandardLifecycles.ts` , `verifiedStandardAmendments.ts` , `verifiedRegulatoryRecords.ts` ) mapping verified BIS records with official source URLs.
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## 4. The 6-Signal Deterministic Matching Engine

The core recommendation engine ( `standardsMatcher.ts` ) calculates a normalized relevance score ( $S \in [0.0, 1.0]$ ) using a weighted linear combination of six independent domain signals:

$$S = \sum_{i=1}^6 w_i \cdot s_i$$

### Factor Weight Distribution:

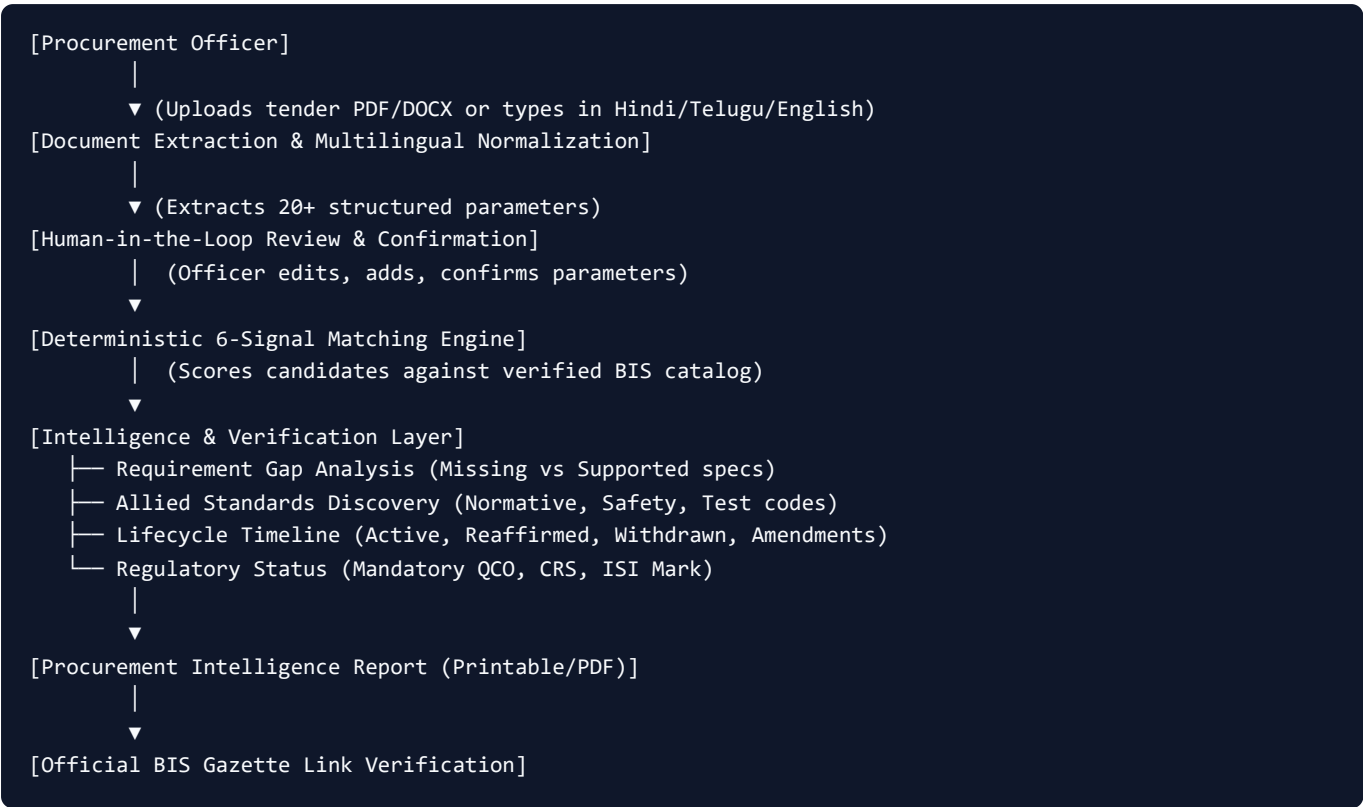
| Factor                  | Weight (\$w_i\$) | Description & Matching Criteria                                                                                               |
|-------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Product / Category      | 30% (0.30)       | Evaluates direct taxonomy alignment between the confirmed product category and the standard's primary product classification. |
| Keywords, Title & Scope | 25% (0.25)       | Matches domain terminology against the standard's official Title and published BIS Scope statement.                           |
| Application Domain Fit  | 15% (0.15)       | Evaluates environment/installation suitability (e.g., Highway/Street lighting vs. Industrial vs. Residential).                |
| Environmental Context   | 10% (0.10)       | Matches ambient operating conditions (e.g., outdoor, high-temperature, corrosive/marine, dust ingress).                       |
| Technical Parameters    | 10% (0.10)       | Evaluates numerical specifications (e.g., wattage, voltage, lumens/watt, conductor core, steel rebar grade).                  |
| Safety & Testing        | 10% (0.10)       | Validates safety protections (surge protection, fire retardance) and mandatory test methods (type tests, routine tests).      |

### Specificity Hierarchy & Penalties

To prevent generic standards from outranking specialized standards:

- **Tiers 1 to 6 Specificity Rating:** Tier 1 represents an exact product-type and application match; Tier 6 represents a broad cross-domain candidate.
- **Product-Form Contradiction Penalties:** When user specifications explicitly demand one application (e.g., *Outdoor Highway Roadway Lighting*), standards explicitly restricted to another form (e.g., *Underground Coal Mining Luminaires*) receive a severe contradiction penalty ( $s_i = 0$  with contradiction status flagged).

## 5. End-to-End Procurement Workflow



## 6. Verification, Traceability & Trust Models

Government tenders require full auditability before the Central Vigilance Commission (CVC) and the Comptroller and Auditor General (CAG). ISutra enforces trust through:

### 1. The 5-Stage Audit Traceability Chain

Every recommended standard includes an unbroken chain of custody:

1. **User Input:** Exact clause or paragraph in the uploaded tender document.
2. **Extracted Requirement:** Structured parameter derived during parsing.
3. **Matching Signal:** Specific factor (e.g., Ingress Protection IP66) evaluated by the algorithm.
4. **Standard Data:** Explicit clause or technical table in the verified BIS standard record.
5. **Official BIS Source:** Direct HTTPS link to `bis.gov.in` or `egazette.gov.in`.

### 2. Four-State Reference Coverage Taxonomy

To avoid false legal claims, ISutra strictly categorizes requirements into:

- `supported` : Confirmed present in the standard.
- `not_supported` : Contradicted or excluded by the standard.
- `not_available` : Standard scope does not govern this specific dimension.

- `needs_verification` : Parameter requires manual inspection in the full BIS publication. *(The system strictly labels this as **Reference Coverage** and never issues a legally binding "compliance certificate").*

## 7. Automated Test Suite & Quality Assurance

ISutra is accompanied by six automated test suites containing **over 280 passing test assertions** (0 failures):

| Test Suite File                                        | Focus Area                                              | Assertions Verified |
|--------------------------------------------------------|---------------------------------------------------------|---------------------|
| <code>backend/test_sih26108_comprehensive.mjs</code>   | Full capability suite (Tests A through T)               | 120+                |
| <code>backend/test_multilingual.mjs</code>             | Hindi, Telugu, and English script handling              | 35+                 |
| <code>backend/test_regulatory_certification.mjs</code> | Mandatory QCO, CRS, and Hallmarking logic               | 30+                 |
| <code>backend/test_allied_standards.mjs</code>         | Normative references and test method resolution         | 25+                 |
| <code>backend/test_phaseF.mjs</code>                   | Real PDF/DOCX ingestion and scanned doc handling        | 30+                 |
| <code>backend/test_differentiation.mjs</code>          | "Why This Standard?" explainability and scoring weights | 40+                 |

## 8. Deployment Specifications

- Hosting Environment:** Cloud-native deployment on **Vercel** with Express serverless functions.
- Hardware Efficiency: Zero specialized GPU dependency.** The entire deterministic scoring engine, regex parsers, and document extractors run on standard low-cost CPU instances (\$ < 512\$ MB memory footprint).
- High Availability:** Fallback architecture ensures that if MongoDB or external AI APIs are unreachable, the system gracefully operates in offline deterministic mode with zero downtime.

Authored by Team Core Coders / Zero Coders for Smart India Hackathon (SIH) 2026 — Problem Statement SIH26108.